Design Brief – Review 2

0. Header

Project code	[Project code from the registration form]
Project name	Lock.R – Smart Locker System with Drone Delivery
Team	Nguyễn Quốc Bảo Huy (SE180211, leader) · Trương Nguyễn Thái Bình (SE182370) · Lê Thị Yến Vi (SE182544) · Trần Kim Nhã (SE182290)
Supervisor	Đặng Ngọc Minh Đức (DucDNM) · Đỗ Hữu Tiến (TienDH)
Repository link (tag review-2)	https://github.com/LockR-Tech (backend, frontend, mobile, iot, docs) – tag review-2 to be created on all five repos
Version	1.0
Date	08/11/2026

1. Scope

1.1 Registration Changes

Section	Current	Proposed	Reason
[Section]	[Current text]	[Proposed text]	[Reason – details in BM03, or "No changes"]

1.2 Out of Scope

Item	Reason
Beyond-visual-line-of-sight commercial drone flights	Needs aviation permits; demo flights run in a controlled area with one drone.
Payment gateways other than SePay VietQR (MoMo, VNPay, cards)	One reconciled payment path; VietQR works with every Vietnamese bank app (BR-22).
Publishing on Google Play / App Store	Pilot ships as an Android APK and mobile web.
More than one physical cabinet	One real cabinet (CAB-TU01, 7 boxes); other lockers run the same controller software simulated.

2. System at a Glance

Lock.R lets people send and store parcels without meeting in person. A customer books a box in a smart locker in the mobile app, pays by VietQR bank transfer and drops the parcel; the receiver opens the box with a 6-digit PIN or QR at the kiosk or over Bluetooth from the phone. Boxes can be rented by the hour, and parcels can travel between lockers by drone. Users are customers and receivers (mobile app, mobile web, kiosk), locker and drone technicians (mobile app) and administrators (admin web). Each locker is a Raspberry Pi that drives relays, door sensors and a sliding landing lid and talks to the backend over MQTT; the backend is a set of Spring Boot microservices behind an API gateway, with an AI assistant. Core value: 24/7 contact-free hand-over in which every box state and payment is traceable. Architecture: SDD Figure 1.1.

3. Key Design Decisions

ID	Decision	Options considered	Chosen option and rationale	Owner	Related FR / diagram
D1	Backend structure	Monolith; microservices	Microservices: 10 Spring Boot services, one database each (Flyway), gateway with JWT and roles, RabbitMQ events; members work per domain and failures stay local.	Huy	SDD Fig. 1.1
D2	Locker control and box opening	Arduino/RS485; vendor controller; Pi on GPIO	Pi 5 on GPIO with a versioned MQTT contract (ACL per device, resend outbox). Orders change only on the cabinet's report; Bluetooth tokens (Ed25519, 120 s, single use) are checked offline on the Pi.	Huy	UC17, UC35; BR-09, BR-25
D3	Payment method	MoMo / VNPay; wallet; SePay VietQR	SePay VietQR only: PAID only when the webhook amount covers the total; duplicate webhooks ignored; no merchant contract, works with every bank app.	Bình	UC08; BR-02, BR-22
D4	Drone delivery orchestration	Auto nearest-drone dispatch; technician accepts offer	Technician accepts, completes a loading checklist and launches; one delivery state machine fed by MQTT telemetry; a person is responsible for every flight.	Vi	UC10, UC32; BR-10, BR-11
D5	Box incident handling	Complaint to admin; box FAULT at once; ticket with SLA	Report puts the box in PENDING_CHECK; a technician on shift claims a ticket (SLA 4 h, max 2 extensions); the scheduler escalates. No box is locked on a false report.	Nhã	UC13, UC28; BR-19, BR-24

4. Top Risks

ID	Risk	Impact	How we verify or mitigate	Owner
R1	Cabinet reports a wrong door state (sensor misread, door stays shut)	Box shown empty while holding a parcel; false JAMMED	20 opens per box on CAB-TU01 with sensor log; hourly box/order reconciliation; emergency open	Huy
R2	Real drone flight cannot be shown reliably (weather, battery, permits)	Drone workflow not shown end to end	Weather check blocks bookings; indoor mode lands on the lid by ArUco with the same events; recorded flight as backup	Vi
R3	External services throttle or delay (Firebase SMS, SePay webhook)	Sign-in fails or orders stay UNPAID in the demo	Firebase test numbers, email OTP fallback; payment polling; service check script before demos	Bình

5. Known Gaps

ID	Gap	Location	Plan	Owner	Due
G1	SRS keeps the customer wallet; the app removed it (SePay only)	SRS UC08, UC12, §3.2.24–26, BR-16	Remove wallet from SRS	Bình	15/11/2026
G2	Opening an order's box over Bluetooth (QR/OTP fallback) is built but not specified	SRS UC16; SDD §3; backend #137	Add use case, rule, sequence diagram	Huy	15/11/2026
G3	State machines and activity diagrams are not yet in Report 3/4	docs/diagrams/src, srs-assets/drawio	Insert 4 + 4 diagrams	Nhã	15/11/2026
G4	No separate staging; demo seed script is outside the repo	Report 4; seed script	Declare staging, move script to repo	Huy	22/11/2026
G5	Report 5 test cases not written; automated tests not mapped to TC IDs	Report 5; test suites	Write Report 5 with §6 TC IDs	Nhã	22/11/2026

ID	Gap	Location	Plan	Owner	Due
G6	[High findings Kept / Planned in BM02]	BM02			

6. Proposed Mandatory Features

ID	Feature	Actor	Acceptance criteria (testable)	Test	Status
F-01	Sign in, admin 2FA	User, Admin	Wrong password: MSG06, no token; admin needs a 6-digit email OTP valid 300 s.	TC-F01-01..03	Done
F-02	Create send order	Customer	Only AVAILABLE boxes of an online locker; box RESERVED; unpaid after 15 min: cancelled, box freed.	TC-F02-01..03	Done
F-03	VietQR payment	Customer	PAID only when received >= total; a duplicate webhook counts once; partial shows the remainder.	TC-F03-01..03	Done
F-04	Drop parcel, share code	Customer	Box opens only if PAID; drop gives STORING / OCCUPIED and a 6-digit PIN to the receiver.	TC-F04-01..03	Done
F-05	Pick up at kiosk	Receiver	Right PIN/QR opens and completes the order; 5 wrong PINs lock the box 15 min (MSG22).	TC-F05-01..03	Done
F-06	Rent box by hours	Customer	1–720 h; multi-use PIN until the end; extension <= 24 h; ending frees the box.	TC-F06-01..03	Done
F-07	Open box by Bluetooth	Customer, Receiver	Token for the order's owner/receiver only, 120 s, single use; state changes once on the Pi report; failure offers QR/OTP.	TC-F07-01..03	In progress
F-08	Overdue fee, expiry	Customer, Scheduler	500 VND/h, max 50,000 VND and 50 % of price; opens only after paying; > 72 h: EXPIRED.	TC-F08-01..03	Done
F-09	Cabinet commands	Locker Controller	No ack in 20 s: TIMEOUT, never success; relay ran but door closed: JAMMED; events resent.	TC-F09-01..03	Done
F-10	Book drone delivery	Customer	Refused: no free DRONE box (MSG13), > 5,000 g, > 5,000 m, unsafe weather or > 3 open orders.	TC-F10-01..03	Done
F-11	Dispatch, load, launch	Drone Technician	Accept only with an idle drone >= 20 % battery; launch needs the checklist; + 50 g re-prices.	TC-F11-01..03	In progress
F-12	Track drone, pickup ready	Customer, Receiver	Stage updates reach the app in 2 s; on arrival the lid opens and the order is STORING with a PIN.	TC-F12-01..03	In progress
F-13	Report box incident	Customer	Title and 1–5 photos required; box PENDING_CHECK, no new orders, overdue fee stops.	TC-F13-01..03	Done
F-14	Ticket within SLA	Locker Tech., Admin	Claim only on shift; SLA 4 h, max 2 extensions with reason; 3 overdue tickets block claims.	TC-F14-01..03	Done
F-15	Parcel drop and recovery	Drone/Locker Tech., Admin	Camera evidence required; recovery SLA 4 h; 60 % compensation needs admin and customer approval.	TC-F15-01..03	In progress
F-16	AI assistant	User	Answers cite sources; order questions see only own orders; reply within 10 s.	TC-F16-01..03	Done

7. Non-Functional Requirements

ID	Quality attribute	Metric	Threshold	Measurement method
N1	Performance	p95 API response time	Read <= 1 s, write <= 2 s at 100 sessions	k6 on staging, 10 min
N2	Responsiveness	PIN / "Open box" to door-open result	<= 5 s in 95 % of 50 opens	Timed run on CAB-TU01 + logs
N3	Realtime	Order status / drone stage delay to app	p95 <= 2 s	Event vs receive time, 30 events
N4	Payment accuracy	Double-counted replayed webhooks	0 of 100	Replay test in CI
N5	Security	PIN lockout; Bluetooth token reuse	Lock after 5 wrong PINs; 100 % reused tokens rejected	Automated tests in CI
N6	Availability	Monthly API uptime	>= 99 %	Health check every 5 min

8. Ownership Map

Member	Role	Owned parts (2–3)	Main workflow owned	Key decisions / PRs
Nguyễn Quốc Bảo Huy	Leader; backend, IoT, DevOps	1) Architecture, gateway/auth, CI/CD, VM; 2) Locker controller, MQTT, Bluetooth open; 3) AI assistant	WF-02	D1, D2; iot-service, iot repo
Trương Nguyễn Thái Bình	Mobile – customer, payment	1) Send / rent / order screens; 2) SePay VietQR payment; 3) Profile, notifications	WF-01	D3; payment-service, mobile
Lê Thị Yến Vi	Drone delivery, admin reports	1) Drone booking and tracking; 2) Admin dashboard, revenue; 3) Loyalty	WF-03	D4; order-service (drone)
Trần Kim Nhã	Maintenance, incidents, QA	1) Locker tickets, SLA, inspections; 2) Parcel drop recovery; 3) Test cases, CI tests	WF-04	D5; locker-service

Appendix A – Defense 1.1 Targets

A1. Main Workflows

ID	Workflow	Actors	Trigger	Business outcome	Features	Entity states	Owner
WF-01	Send parcel via locker, receiver picks up	Customer, Receiver, Locker Controller	Customer creates a SEND order	Parcel handed over; order COMPLETED	F-01–F-05, F-09, F-16	Order INITIALIZED → STORING → COMPLETED; Box AVAILABLE → RESERVED → OCCUPIED → AVAILABLE	Bình
WF-02	Rent a box by hours	Customer, Locker Controller, System Scheduler	Customer rents a box	Rental ended, or EXPIRED after the overdue fee	F-03, F-06–F-09	Order INITIALIZED → STORING → COMPLETED / EXPIRED; Box RESERVED → OCCUPIED → AVAILABLE	Huy
WF-03	Drone delivery between lockers	Customer, Drone Technician, Receiver, Locker Controller	Customer books a drone delivery	Parcel in the receiver locker and picked up	F-03, F-05, F-10–F-12, F-15	Order AWAITING_DISPATCH → STORING → COMPLETED; Stage ACCEPTED → … → READY_FOR_PICKUP; Drone IDLE → RESERVED → IN_FLIGHT → IDLE	Vi
WF-04	Box incident to repair	Customer, Locker Technician, Admin	Customer reports a faulty box	Box back in service or locked	F-13, F-14	Box → PENDING_CHECK → FAULT → AVAILABLE; Ticket OPEN → IN_PROGRESS → RESOLVED / REJECTED	Nhã

A2. Demo Scenarios

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-01	WF-01	Main	CAB-TU01 has a free Standard box; customer account; receiver phone	Create order → pay by VietQR → open box and drop → receiver enters PIN at kiosk	Order COMPLETED, box AVAILABLE	Bình
DS-02	WF-01	Exception	Order DS-01 in STORING	Enter a wrong PIN 5 times at the kiosk	Box locked 15 min (MSG22); order stays STORING	Bình
DS-03	WF-02	Main	Customer near CAB-TU01 with Bluetooth on	Rent 2 h → pay → "Open box" (Bluetooth) → reopen → end rental	Box opens over Bluetooth; order COMPLETED, box AVAILABLE	Huy
DS-04	WF-02	Exception	Bluetooth off; a second rental past its end time	Tap "Open box" → choose OTP at kiosk; open the overdue rental	App offers QR/OTP; overdue box opens only after the fee is paid	Huy

ID	Workflow	Type	Preconditions and data	Main steps	Expected result	Presenter
DS-05	WF-03	Main	Two lockers with landing pads; idle drone >= 20 % battery; safe weather	Book → pay → technician accepts, loads, launches → arrival → receiver picks up	Lid opens; order STORING with PIN, then COMPLETED	Vi
DS-06	WF-03	Exception	Receiver locker with no free drone box; parcel of 6,000 g	Book a drone delivery	Booking refused (MSG13 / weight limit)	Vi
DS-07	WF-04	Main	Occupied box; locker technician on shift	Customer reports with a photo → technician claims → resolves	Box PENDING_CHECK → back in service; ticket RESOLVED	Nhã
DS-08	WF-04	Exception	Ticket already extended twice; technician with 3 overdue tickets	Extend the SLA a third time; claim a new ticket	Both refused with a message	Nhã

A3. Functional Test Case Target

Source	Count	Minimum per item	Subtotal
Mandatory features	16	3	48
Main workflows (end-to-end)	4	1	4
State transitions of main entities	44	1	44
Agreed NFRs	6	1	6
Committed total			102

Automated in CI: at least 50% of the committed total.